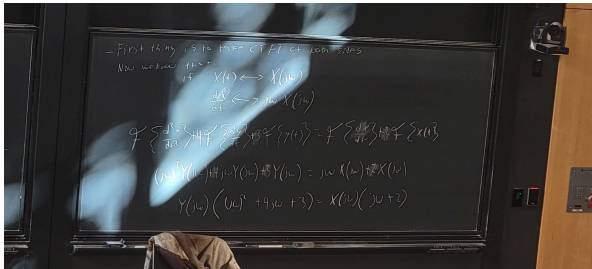
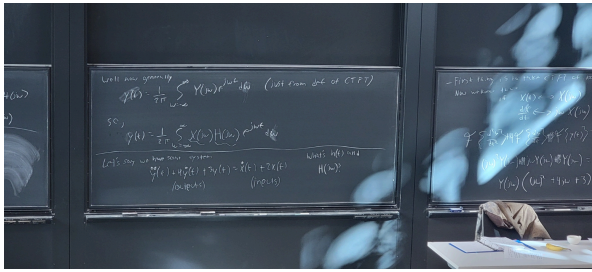


14 Convolution in Fourier

Continuous



Then we get $H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{j\omega+2}{(j\omega)^2+4j\omega+3}$

The inverse Fourier transform of this is $h(t)$. This is most easily done with the Fourier transform pairs that we derived.

We use partial fractions to separate out

$$H(j\omega) = \frac{\frac{1}{2}}{j\omega + 1} + \frac{\frac{1}{2}}{j\omega + 3}$$

Recall

$$e^{-\alpha t}u(t) \leftrightarrow \frac{1}{\alpha + j\omega}$$

$$\text{so } h(t) = \frac{1}{2}e^{-t}u(t) + \frac{1}{2}e^{-3t}u(t)$$

Lets take an arbitrary input $x(t) = e^{-4t}$. The Fourier transform is $X(j\omega) = \frac{1}{4+j\omega}$

Then we have

$$Y(j\omega) = H(j\omega)X(j\omega)$$

and then go back to the time domain with the Fourier pairs.

One more interesting property:

Multiplication and convolution are DUALS

$$\begin{aligned} x(t) y(t) &\leftrightarrow X(j\omega) Y(j\omega) \\ x(t) y(t) &\leftrightarrow \frac{1}{2\pi} X(j\omega) Y(j\omega) \end{aligned}$$

Note that if the Fourier domain is a bunch of delta functions that are far enough apart, then we just get multiple copies of the original function.

Discrete

Lets take an aperiodic signal x

We truncate it to only exist over one period T , and zero elsewhere, then make it a periodic function by making copies at a period T

call this $\tilde{x}$ with period T and $\omega_0 = \frac{2\pi}{T}$.

We now have the Fourier series representation

$$a_k = \frac{1}{T} \tilde{x} e^{-jk(\frac{2\pi}{T})} \text{ and } \tilde{x} = \sum_{k=-\infty}^{\infty} a_k e^{jk(\frac{2\pi}{T})}.$$

This is the same as

$$a_k = \frac{1}{T} \sum_{n=-\infty}^{\infty} x e^{-jn(\frac{2\pi}{T})}$$

We define

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x e^{-jn\omega}, \text{ so}$$
$$a_k = \frac{1}{T} X(e^{jk\omega_0})$$